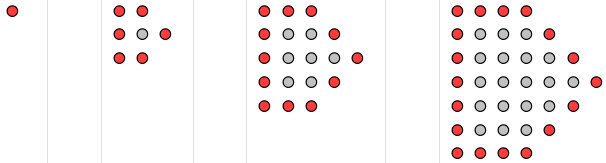
In mathematics and combinatorics, a **centered hexagonal number**, or **centered hexagon number**,1([https://en.wikipedia.org/wiki/Centered_hexagonal_number#cite_note-1](https://en.wikipedia.org/wiki/Centered_hexagonal_number#cite_note-1))[[2]](https://en.wikipedia.org/wiki/Centered_hexagonal_number#cite_note-Deza-2) is a centered figurate number that represents a hexagon with a dot in the center and all other dots surrounding the center dot in a hexagonal lattice. The following figures illustrate this arrangement for the first four centered hexagonal numbers:

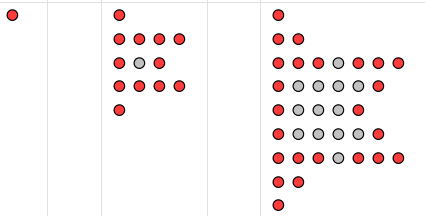
| **1** | | **7** | | **19** | | **37** |
|---|---|---|---|---|---|---|
| +1 | | +6 | | +12 | | +18 |
|  | | | | | | |

Centered hexagonal numbers should not be confused with cornered hexagonal numbers, which are figurate numbers in which the associated hexagons share a vertex.

The sequence of hexagonal numbers starts out as follows (sequence A003215 in the OEIS):

1, 7, 19, 37, 61, 91, 127, 169, 217, 271, 331, 397, 469, 547, 631, 721, 817, 919.

The numbers are also called **centered dodecagonal numbers** because star numbers are centered polygonal numbers with a twelve-sided shape.

| **1** | | **13** | | **37** |
|---|---|---|---|---|
|  | | | | |

The *n_th star number is given by the formula _Sn* = 6*n*(*n* − 1) + 1. The first 45 star numbers are 1, 13, 37, 73, 121, 181, 253, 337, 433, 541, 661, 793, 937, 1093, 1261, 1441, 1633, 1837, 2053, 2281, 2521, 2773, 3037, 3313, 3601, 3901, 4213, 4537, 4873, 5221, 5581, 5953, 6337, 6733, 7141, 7561, 7993, 8437, 8893, 9361, 9841, 10333, 10837, 11353, and 11881. (sequence A003154 in the OEIS)

The digital root of a star number is always 1 or 4, and progresses in the sequence 1, 4, 1. The last two digits of a star number in base 10 are always 01, 13, 21, 33, 37, 41, 53, 61, 73, 81, or 93.

Unique among the star numbers is 35113, since its prime factors (i.e., 13, 37 and 73) are also consecutive star numbers.

---

There was one design decision. The delta law:

rotl(x,1) XOR rotl(x,3) XOR rotr(x,2) XOR C

Four choices inside that: rotations (not shifts, so no bits are lost), XOR (reversible), a constant (breaks the zero fixed point), mask to width (keeps state bounded).

That is the entire design.

Everything else was computed by the math:

Period = 8 → property of that specific law on 16-bit space

Prime 73 → smallest prime whose decimal repeats with period 8

Block B = [0,1,3,6,9,8,6,3] → the digits of 1/73

W = 36 → sum of B

Orbit/offset recovery → just divmod(position, 36)

Nobody chose 73. The law has period 8. The smallest prime with decimal period 8 is 73. The system found it.

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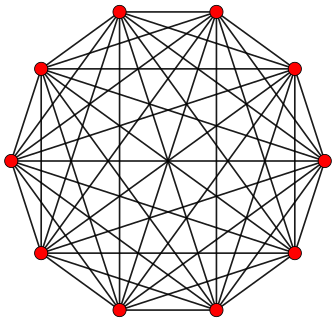
| | |
|---|---|
| ? | This article includes a list of general references, but **it lacks sufficient corresponding inline citations**. Please help to improve this article by introducing more precise citations. *(August 2025) (Learn how and when to remove this message)* |

| **Regular decayotton (9-simplex)** | |
|---|---|
|  Orthogonal projection inside Petrie polygon | |
| Type | Regular 9-polytope |
| Family | simplex |
| Schläfli symbol | {3,3,3,3,3,3,3,3} |
| Coxeter-Dynkin diagram |  |
| 8-faces | 10 8-simplex |
| 7-faces | 45 7-simplex |
| 6-faces | 120 6-simplex |
| 5-faces | 210 5-simplex |
| 4-faces | 252 5-cell |
| Cells | 210 tetrahedron |
| Faces | 120 triangle |
| Edges | 45 |
| Vertices | 10 |
| Vertex figure | 8-simplex |
| Petrie polygon | decagon |
| Coxeter group | A9 [3,3,3,3,3,3,3,3] |
| Dual | Self-dual |
| Properties | convex |

In geometry, a 9-simplex is a self-dual regular 9-polytope. It has 10 vertices, 45 edges, 120 triangle faces, 210 tetrahedral cells, 252 5-cell 4-faces, 210 5-simplex 5-faces, 120 6-simplex 6-faces, 45 7-simplex 7-faces, and 10 8-simplex 8-faces. Its dihedral angle is cos−1(1/9), or approximately 83.62°.

It can also be called a **decayotton**, or **deca-9-tope**, as a 10-facetted polytope in 9-dimensions. The name *decayotton* is derived from *deca* for ten facets in Greek and yotta (a variation of "oct" for eight), having 8-dimensional facets, and *-on*.
Jonathan Bowers gives it acronym **day**.1(https://en.wikipedia.org/wiki/9-simplex#cite_note-1)

## Coordinates

The Cartesian coordinates of the vertices of an origin-centered regular decayotton having edge length 2 are:

(1/45, 1/6, 1/28, 1/21, 1/15, 1/10, 1/6, 1/3, ±1) $\left(\sqrt{1/45},\ 1/6,\ \sqrt{1/28},\ \sqrt{1/21},\ \sqrt{1/15},\ \sqrt{1/10},\ \sqrt{1/6},\ \sqrt{1/3},\ \pm 1\right)$

(1/45, 1/6, 1/28, 1/21, 1/15, 1/10, 1/6, −21/3, 0) $\left(\sqrt{1/45},\ 1/6,\ \sqrt{1/28},\ \sqrt{1/21},\ \sqrt{1/15},\ \sqrt{1/10},\ \sqrt{1/6},\ -2\sqrt{1/3},\ 0\right)$

(1/45, 1/6, 1/28, 1/21, 1/15, 1/10, −3/2, 0, 0) $\left(\sqrt{1/45},\ 1/6,\ \sqrt{1/28},\ \sqrt{1/21},\ \sqrt{1/15},\ \sqrt{1/10},\ -\sqrt{3/2},\ 0,\ 0\right)$

(1/45, 1/6, 1/28, 1/21, 1/15, −22/5, 0, 0, 0) $\left(\sqrt{1/45},\ 1/6,\ \sqrt{1/28},\ \sqrt{1/21},\ \sqrt{1/15},\ -2\sqrt{2/5},\ 0,\ 0,\ 0\right)$

(1/45, 1/6, 1/28, 1/21, −5/3, 0, 0, 0, 0) $\left(\sqrt{1/45},\ 1/6,\ \sqrt{1/28},\ \sqrt{1/21},\ -\sqrt{5/3},\ 0,\ 0,\ 0,\ 0\right)$

(1/45, 1/6, 1/28, −12/7, 0, 0, 0, 0, 0) $\left(\sqrt{1/45},\ 1/6,\ \sqrt{1/28},\ -\sqrt{12/7},\ 0,\ 0,\ 0,\ 0,\ 0\right)$

(1/45, 1/6, −7/4, 0, 0, 0, 0, 0, 0) $\left(\sqrt{1/45},\ 1/6,\ -\sqrt{7/4},\ 0,\ 0,\ 0,\ 0,\ 0,\ 0\right)$

(1/45, −4/3, 0, 0, 0, 0, 0, 0, 0) $\left(\sqrt{1/45},\ -4/3,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0\right)$

(−31/5, 0, 0, 0, 0, 0, 0, 0, 0) $\left(-3\sqrt{1/5},\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0\right)$

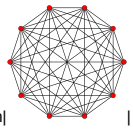
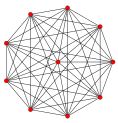
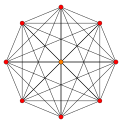
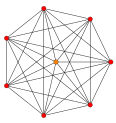
More simply, the vertices of the *9-simplex* can be positioned in 10-space as permutations of (0,0,0,0,0,0,0,0,0,1). These are the vertices of one Facet of the 10-orthoplex.

## Images

| | | | | |
|---|---|---|---|---|

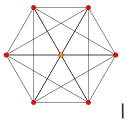
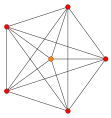
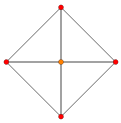
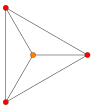
orthographic projections

|Ak Coxeter plane|A9|A8|A7|A6|

|Graph|  |  |  |  |

|Dihedral symmetry|[10]|[9]|[8]|[7]|

|Ak Coxeter plane|A5|A4|A3|A2|

|Graph|  |  |  |  |

|Dihedral symmetry|[6]|[5]|[4]|[3]|